Lab Manual 5

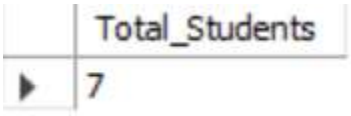
Part A – Aggregate Functions

1.

SQL

```
SELECT COUNT(*) AS Total_Students FROM Student;
```

OUTPUT



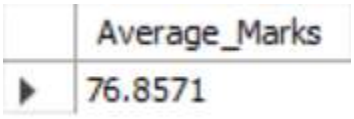
	Total_Students
▶	7

2.

SQL

```
SELECT AVG(Marks) AS Average_Marks FROM Student;
```

OUTPUT



	Average_Marks
▶	76.8571

3.

SQL

```
SELECT MAX(Marks) AS Max_Marks, MIN(Marks) AS Min_Marks FROM
Student;
```

OUTPUT

	Max_Marks	Min_Marks
▶	92	55

4.

SQL

```
SELECT SUM(Marks) AS Total_Marks FROM Student;
```

OUTPUT

	Total_Marks
▶	538

Part B – GROUP BY Clause

5.

SQL

```
SELECT Department, COUNT(*) AS Student_Count
FROM Student
GROUP BY Department;
```

OUTPUT

	Department	Student_Count
▶	CSE	3
	ECE	2
	ME	2

6.

SQL

```
SELECT Department, AVG(Marks) AS Avg_Marks
FROM Student
GROUP BY Department;
```

OUTPUT

	Department	Avg_Marks
▶	CSE	84.3333
	ECE	78.5000
	ME	64.0000

7.

SQL

```
SELECT Department, MAX(Marks) AS Max_Marks
FROM Student
GROUP BY Department;
```

OUTPUT

	Department	Max_Marks
▶	CSE	90
	ECE	92
	ME	73

Part C – HAVING Clause

8.

SQL

```
SELECT Department, AVG(Marks) AS Avg_Marks
FROM Student
GROUP BY Department
HAVING AVG(Marks) > 70;
```

OUTPUT

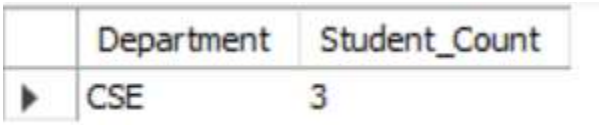
	Department	Avg_Marks
▶	CSE	84.3333
	ECE	78.5000

9.

SQL

```
SELECT Department, COUNT(*) AS Student_Count
FROM Student
GROUP BY Department
HAVING COUNT(*) > 2;
```

OUTPUT



	Department	Student_Count
▶	CSE	3

10.

SQL

```
SELECT Department, MAX(Marks) AS Max_Marks
FROM Student
GROUP BY Department
HAVING MAX(Marks) > 90;
```

OUTPUT



	Department	Max_Marks
▶	ECE	92

Part D – Combined Queries

11.

SQL

```
SELECT Department, SUM(Marks) AS Total_Marks
FROM Student
GROUP BY Department
HAVING SUM(Marks) > 200;
```

OUTPUT

	Department	Total_Marks
▶	CSE	253

12.

SQL

```
SELECT Department, AVG(Marks) AS Avg_Marks
FROM Student
GROUP BY Department
HAVING AVG(Marks) BETWEEN 60 AND 80;
```

OUTPUT

	Department	Avg_Marks
▶	ECE	78.5000
	ME	64.0000